A Computer-Assisted Bound for 2-Weakly Compatible Split Systems

A standalone exposition of the strict-exponent argument

Research note

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Abstract

We give the complete mathematical argument for the computer-assisted bound $|\mathcal{S}| = O(n^{5/2-1/1736954})$. Quadratic-cost deletions remove small sides, non-path second links, degree-six squares, cubes with degree-six bottom and top, and regular degree-five cubes. In the remaining family a finite local computation puts positive charge within distance two of every degree-six vertex. A second local result controls the degree-five bulk. Together they give a strict saving in the edge coefficient, which transfers to the size exponent by standard reduction. This document contains the required lemmas and their proofs, rather than the chronological research record. The elementary facts explicitly attributed to Wu and Grünewald are used from their paper. Finite verification programs and data are computational dependencies, listed below; they are not embedded in the PDF. The computations use Z3, without an independently checked formal UNSAT certificate. The coefficient-two conjecture is not proved.

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1 Statement and proof structure

Theorem 1.1 (Computer-assisted strict improvement)**.** *Put*

$$\alpha = \frac{5}{2} - \frac{1}{1736954}.$$

There is an absolute constant C such that, for every 2-weakly compatible split system on n taxa and every root,

$$|E| \le \alpha|V| + Cn^2.$$

Consequently its size is $O(n^\alpha)$, with an absolute implicit constant. In particular $\alpha < 5/2$.

Every constant below is absolute. Put $\delta = 1/1736954$ and $\alpha = 5/2 - \delta$. We prove the edge inequality first. Its implication for the size bound is proved explicitly in the last section.

The proof has the following quantitative steps. After deleting $O(n^2)$ vertices and incident edges, write V^*, E^* for the remaining graph, H for its number of degree-six vertices, and p for its number of positive-charge vertices. We establish

$$H \le 37p, \qquad V^* \le 938p + 23447H, \qquad 5V^* - 2E^* \ge p.$$

Thus $V^* \le 868477p$ and $E^* \le \alpha V^*$. All pruning statements are proved for the original unrestricted class. Only the original class is used in the final recurrence; no closure of the pruned class under taxon deletion is assumed.

The finite parts are the cube regularity theorem, the regular degree-five ball theorem, the two foreign-face classifications, and the final charge-proximity theorem. Their normalizations and necessary constraints are given in the corresponding sections. The remaining counting and recurrence arguments are by hand.

2 Setup and dependencies

Let X be a set of n taxa. A split is an unordered bipartition $A \mid (X \setminus A)$, including the empty split. Fix $r \in X$ and represent each split of $\mathcal{S}$ by its side avoiding r. Let $V = \mathcal{F} \subseteq 2^{X\setminus\{r\}}$ be the resulting family. The directed graph has an edge $A \to A + x$ whenever both sets belong to V, where $A + x = A \cup \{x\}$ and $A - x = A \setminus \{x\}$. Write d^+, d^-, d for outdegree, indegree, and total degree. Throughout, E denotes this graph's edge set.

The system is *2-weakly compatible* if there do not exist six distinct taxa $h_1, h_2, q_1, q_2, q_3, q_4$ and four splits which, after restriction and independent choices of sides, have the form

$$\{h_1, h_2, q_i\} \mid \{q_j : j \ne i\}, \qquad i = 1, 2, 3, 4. \tag{2.1}$$

We call h_1, h_2 the anchors and q_i the private taxon of split i. All witnesses below specify four sets, the anchors, the independent choices of sides, and the four private taxa. Thus each can be checked directly against (2.1).

The established bound in [1] is $|E| \le (5/2)|V| + (25/24)n^2$.

We use the following facts from Lemmas 1–3 and the proof of Theorem 2 of [1]. We state the dependencies explicitly; we do not claim them as new results.

(i) If both sides of a split have size at least two, then $d^+, d^- \le 3$. A 2-split has degree at most five. Each empty or trivial split has degree at most $n - 1$, and there are at most $n + 1$ such splits.

(ii) A degree-six vertex A is *hard* when its neighbors can be labelled $A - a_i, A + b_i$ for $1 \le i \le 3$ so that all three matching swaps $P_i = A - a_i + b_i$ are vertices. A matching swap P_i is a

partner when no off-diagonal swap in its row or column is present. Every hard vertex has a partner. For a partner $A - a + b$,

$$d^+(A - a) = d^-(A + b) = 2. \tag{2.2}$$

No vertex is a partner of two distinct hard vertices.

(iii) A nontrivial, non-hard degree-six vertex has a neighbor whose unique outgoing or incoming neighbor, respectively, is that vertex. Thus it receives at least one unit under the first rule in Section 4.

For context, (iii) follows by examining the bipartite swap graph on $\{a_1, a_2, a_3\}$ and $\{b_1, b_2, b_3\}$. Compatibility excludes degree three and two vertex-disjoint paths of length two. If it has no isolated vertex, it has a perfect matching and A is hard. An isolated row or column supplies the indicated neighbor. Additional successors or predecessors not represented in this swap graph are excluded by (2.1).

Call a vertex *balanced* if

$$5 \le |A| \le n - 5. \tag{2.3}$$

For a balanced hard vertex, put $B = \{b_1, b_2, b_3\}$. Its six edge labels are distinct, and balance gives

$$|A \setminus \{a_1, a_2, a_3\}| \ge 2, \qquad |X \setminus (A \cup B)| \ge 2. \tag{2.4}$$

For a taxon x, the derivative $\partial_x \mathcal{S}$ consists of the splits $A \mid B$ of $X - x$ for which both $(A+x) \mid B$ and $A \mid (B+x)$ belong to $\mathcal{S}$. A second derivative is defined by iteration: its members have all four lifts on its two new taxa. For nonroot x, y, a rooted old side B is in the second link exactly when $B, B + x, B + y, B + x + y$ all occur. An *ordinary compatible* split system has no crossing pair: for each two splits, at least one of their four side intersections is empty. This stronger condition will be applied to second links. Compatibility of the original family always refers to the 2-weak condition (2.1).

3 A low-degree partner always exists

Fix a balanced hard vertex A and matching swaps $P_i = A - a_i + b_i$. If $d^+(P_i) = 3$, one outgoing neighbor is $A + b_i$. Write the others as $P_i + c_i, P_i + e_i$, and set $C_i = \{c_i, e_i\}$. The two elements of C_i are distinct, lie outside A, and differ from b_i.

Lemma 3.1. *The three matching swaps cannot all have outdegree three.*

Proof. Suppose they do. We establish three constraints on the lists C_i.

First, each C_i contains another b-label. If $C_i \cap B = \varnothing$, let $\{i, j, k\} = \{1, 2, 3\}$ and consider

$$P_j, \quad P_k, \quad P_i + c_i, \quad P_i + e_i.$$

Take two anchors in $A \setminus \{a_1, a_2, a_3\}$, retain all displayed sides, and use private taxa b_j, b_k, c_i, e_i, in that order. This is (2.1). Hence

$$C_i \cap (B \setminus \{b_i\}) \ne \varnothing. \tag{3.1}$$

Second, each b_i appears in at most one other list. If $b_i \in C_j \cap C_k$ for the other two indices, use

$$A, \quad P_i, \quad P_j + b_i, \quad P_k + b_i.$$

Choose two anchors outside $A \cup B$ and complement all four sides. The respective private taxa are b_i, a_i, a_j, a_k, a contradiction.

Third, occurrences are reciprocal. Suppose $b_j \in C_i$ but $b_i \notin C_j = \{c, e\}$. Consider

$$A + b_i, \quad P_i + b_j, \quad P_j + c, \quad P_j + e.$$

The anchors are a_j, b_i. Keep the first two sides and complement the last two. The private taxa are, respectively,

$$a_i, \quad b_j, \quad e, \quad c.$$

These six taxa are distinct: c, e lie outside A and avoid both b_j and, by assumption, b_i. This proves

$$b_j \in C_i \implies b_i \in C_j. \tag{3.2}$$

Now form the loopless digraph on $\{1, 2, 3\}$ with arc $i \to j$ if $b_j \in C_i$. Its outdegrees are positive by (3.1), while its indegrees are at most one by the second constraint. It therefore has exactly three arcs. But (3.2) pairs all arcs into reciprocal pairs, which is impossible for an odd number of arcs. □

Lemma 3.2. *If $d^+(P_1) = d^+(P_2) = 3$, then $d^-(P_3) \leq 2$.*

Proof. The second constraint in the preceding proof applies to the two existing lists C_1, C_2: b_3 cannot belong to both. Relabel $1, 2$ so that $b_3 \notin C_1 = \{c, e\}$.

First $P_3 - a_1$ cannot occur. Otherwise the four sets

$$P_3 - a_1, \quad A, \quad P_1 + c, \quad P_1 + e$$

are forbidden with two anchors in $A \setminus \{a_1, a_3\}$, all sides displayed, and respective private taxa b_3, a_1, c, e.

If $d^-(P_3) = 3$, besides the predecessor $A - a_3$ obtained by deleting b_3, there are predecessors $P_3 - d, P_3 - f$ for distinct $d, f \in A \setminus \{a_3\}$. The preceding paragraph gives $a_1 \notin \{d, f\}$. Now use

$$P_1 + c, \quad P_1 + e, \quad P_3 - d, \quad P_3 - f.$$

Take anchors a_1, b_3, complement the first two sides, and keep the last two. The private taxa are e, c, f, d, respectively. This is forbidden. □

Both lemmas are symmetric under exchanging the two sides of every split and reversing all edges. Such an exchange is used only as a proof symmetry: the witnesses concern unordered splits and do not require a change to the fixed rooted graph.

Corollary 3.3. *At least one of the three matching swaps has total degree at most four.*

Proof. Otherwise every P_i has indegree three or outdegree three, since each directed degree is at most three. Two have the same directed degree three. Reverse sides if necessary and relabel to arrange $d^+(P_1) = d^+(P_2) = 3$. Lemma 3.2 implies $d^-(P_3) \leq 2$. The assumption $d(P_3) \geq 5$ forces $d^+(P_3) = 3$, contrary to Lemma 3.1. □

Theorem 3.4. *Every balanced hard degree-six vertex has a partner of degree at most four.*

Proof. If there are no off-diagonal swaps, all three P_i are partners, so Corollary 3.3 applies.

Otherwise relabel an off-diagonal swap as $A - a_1 + b_2$. It prevents P_1 and P_2 from being partners. Partner existence then implies that P_3 is a partner. We reproduce the short argument for this case, already present in the first case of Lemma 4 of [1].

Suppose $d^+(P_3) = 3$, with extra labels c, e. If neither is b_2, use

$$P_3 + c, \quad P_3 + e, \quad A - a_2 + b_2, \quad A - a_1 + b_2.$$

Choose anchors a_3, b_2, complement the first two sides, and keep the last two. The private taxa are e, c, a_1, a_2, respectively. If an extra label is b_2, instead use

$$A - a_2 + b_2, \quad A - a_1 + b_2, \quad A, \quad P_3 + b_2.$$

Complement all sides and choose anchors b_1, r. The private taxa are a_2, a_1, b_2, a_3. Thus $d^+(P_3) \leq 2$. Reverse sides and relabel the off-diagonal swap to prove $d^-(P_3) \leq 2$. For this reversed witness an element of $A \setminus \{a_1, a_2, a_3\}$ supplies the extra fixed anchor used in place of r. Hence $d(P_3) \leq 4$. □

4 A nonnegative conserved charge

For every vertex set $C_0(A) = 5 - d(A)$. Apply these simultaneous rules in two successive stages. A degree-six recipient in the first rule is required to be nontrivial, equivalently to have three edges in each direction:

1. A vertex with a unique outgoing neighbor sends one unit to that neighbor if it is nontrivial and has degree six. Apply the analogous rule for a unique incoming neighbor. Denote the result by C_1.

2. Each partner of degree at most four of a balanced hard vertex sends one unit to that hard vertex. Denote the result by C_2.

Unlike the paper's rules, the first stage need not target partners, and the second stage need not use high-degree partners that received charge.

Proposition 4.1. *Every balanced vertex has $C_2 \geq 0$. Both rules preserve the total charge.*

Proof. A stage-one donor is adjacent to a nontrivial degree-six vertex, so it cannot be an empty or trivial split. A donor with a unique outgoing or incoming neighbor has degree at most four and initial charge at least one. If it sends in both directions, its degree is two and initial charge is three. Hence every donor remains nonnegative.

A nontrivial, non-hard degree-six vertex receives at least one unit by dependency (iii), and its charge therefore becomes nonnegative. A vertex of degree at most five that does not donate has nonnegative initial charge.

Next let $P = A - a + b$ be a partner of a balanced hard vertex. It sends nothing in stage one: if it has unique outgoing neighbor, that neighbor is $A + b$, whose indegree is two by (2.2), so the neighbor is not a degree-six vertex. The incoming argument is identical. If $d(P) \leq 4$, it therefore has at least one unit before stage two. Partner uniqueness allows it to send to at most one hard vertex. Finally, every balanced hard vertex receives at least one unit by Theorem 3.4, cancelling its initial charge -1. Every transfer subtracts and adds the same quantity, proving conservation. □

5 Swap rectangles force positive charge

Let A be balanced, and suppose that the nine vertices

$$A, \qquad L_i = A - a_i, \qquad U_j = A + b_j, \qquad P_{ij} = A - a_i + b_j \quad (i, j \in \{1, 2\}) \tag{5.1}$$

are present, for distinct labels a_1, a_2, b_1, b_2. We call this a *swap rectangle centered at A*. We will prove that a corner P_{ij} must retain charge. The conclusion does not assume that A or its axial neighbors L_i, U_j have zero charge.

Lemma 5.1. *Among the four corners of a balanced swap rectangle, all corners of outdegree three belong to at most one row, and all corners of indegree three belong to at most one column. In particular, some corner has both directed degrees at most two.*

Proof. Suppose $d^+(P_{ij}) = 3$. Besides U_j, its outgoing neighbors are $P_{ij} + c$ and $P_{ij} + e$, where c, e are distinct elements outside A other than b_j. Let i', j' denote the other row and column. If $b_{j'} \notin \{c, e\}$, the four sets

$$L_{i'}, \quad P_{ij'}, \quad P_{ij} + c, \quad P_{ij} + e$$

are forbidden. Retain the displayed sides, choose two anchors in $A \setminus \{a_1, a_2\}$, and take private taxa $a_i, b_{j'}, c, e$. Consequently an outdegree-three corner in row i forces the vertex

$$Q_i = A - a_i + b_1 + b_2.$$

Both Q_1, Q_2 cannot occur: complement the four sets U_1, U_2, Q_1, Q_2, choose two anchors outside $A \cup \{b_1, b_2\}$, and use private taxa b_2, b_1, a_1, a_2. Balance supplies the anchors. Thus at most one row contains an outdegree-three corner.

Exchanging the sides of all splits exchanges rows with columns and outdegree with indegree. The same argument proves the assertion about columns. Choose a row containing no outdegree-three corner and a column containing no indegree-three corner. Their intersection has both directed degrees at most two, since all directed degrees here are at most three. □

Degree at most four alone is not enough: a vertex might donate its charge to a hard vertex elsewhere. The next lemma excludes exactly this possibility.

Lemma 5.2. *No corner of a balanced swap rectangle is a partner of a hard vertex.*

Proof. By relabelling, suppose $P = P_{11}$ is a partner of a hard vertex H. Write $H = P - d + c$ with $d \in P$ and $c \notin P$; since all displayed sets avoid r, also $c \neq r$. The partner identities give

$$d^+(P - d) = d^-(P + c) = 2. \tag{5.2}$$

But L_1 has outgoing neighbors A, P_{11}, P_{12}, so $d \neq b_1$. Likewise U_1 has incoming neighbors A, P_{11}, P_{21}, so $c \neq a_1$. If $d = a_2$, then $P - d$ has the three distinct outgoing neighbors P_{11}, P_{21}, H, contrary to (5.2). If $c = b_2$, then $P + c$ has the three distinct incoming neighbors P_{11}, P_{12}, H, again a contradiction. Therefore

$$d \in A \setminus \{a_1, a_2\}, \qquad c \notin A \cup \{b_1, b_2, r\}, \qquad H = A - a_1 - d + b_1 + c. \tag{5.3}$$

The successor $H + a_1$ cannot occur. Indeed the four sets

$$A, \quad P_{21}, \quad P_{11}, \quad H + a_1$$

are forbidden after complementing all sides: anchors are b_2, r and private taxa are b_1, a_2, a_1, d.

The successor $H + b_2$ cannot occur either. Use

$$A, \quad P_{21}, \quad P_{22}, \quad H + b_2,$$

retain the first three sides and complement the last. Anchors are a_1, d; private taxa are a_2, b_1, b_2, r.

Since H is hard, it has three outgoing neighbors. One is $H + d = P + c$. Write the others as $H + e, H + f$. The only elements of A missing from H are a_1, d, and a_1 was just excluded. Hence e, f lie outside A. They avoid b_1, c, which are already in H, and b_2, which was also excluded. They are distinct. Now the four sets

$$L_2, \quad P_{12}, \quad H + e, \quad H + f$$

are forbidden with all sides displayed, two anchors in $A \setminus \{a_1, a_2, d\}$, and private taxa a_1, b_2, e, f. The two anchors exist because $|A| \geq 5$. This final contradiction proves the lemma. □

Theorem 5.3. *Every balanced swap rectangle has a corner P with $d^+(P), d^-(P) \le 2$ and $C_2(P) \ge 1$. In particular, its four corners cannot all have zero final charge.*

Proof. Choose P by Lemma 5.1. Its directed degrees are at least one because the axial neighbors in (5.1) are present. Its first-stage loss is at most

$$\mathbf{1}_{d^+(P)=1} + \mathbf{1}_{d^-(P)=1} = (2 - d^+(P)) + (2 - d^-(P)) = 4 - d(P).$$

It follows that $C_1(P) \ge 5 - d(P) - (4 - d(P)) = 1$. By Lemma 5.2, P makes no second-stage donation. Hence $C_2(P) \ge 1$. □

6 An endpoint forces a cube

Write $L_i = A - a_i$, $U_j = A + b_j$, and $P_{ij} = A - a_i + b_j$, with distinct $a_1, a_2 \in A$ and distinct $b_1, b_2, b_3 \notin A$, none equal to the root. Assume A is balanced and all ten vertices

$$A, L_1, L_2, U_1, U_2, U_3, P_{11}, P_{12}, P_{22}, P_{23}$$

are present. Here the swap edges form the path $b_1 - a_1 - b_2 - a_2 - b_3$.

Proposition 6.1. *If $d(U_3), d(P_{23}) \ge 5$, then $A + b_1 + b_3$, $A + b_2 + b_3$, and $P_{23} + b_2$ are present. Consequently the full cube with base L_2 and directions a_2, b_2, b_3 is present.*

Proof. First $d^-(U_3) = 2$. Its known predecessors are A, P_{23}. Another predecessor is $A - d + b_3$ with $d \in A \setminus \{a_2\}$. For $d = a_1$, the four sets $L_2, P_{11}, P_{12}, A - a_1 + b_3$ have two common anchors in $A \setminus \{a_1, a_2\}$ and respective private taxa a_1, b_1, b_2, b_3, a contradiction. For $d \notin \{a_1, a_2\}$, use $P_{11}, P_{12}, A - d + b_3, P_{23}$, complement the first two, and take anchors a_1, b_3 and respective privates b_2, b_1, a_2, d.

Next $d^-(P_{23}) \le 2$. One predecessor is L_2. Deleting a_1 is forbidden by the first witness just used, replacing its fourth set by $P_{23} - a_1$. If two other predecessors $P_{23} - d, P_{23} - e$ existed, then $d, e \in A \setminus \{a_1, a_2\}$. Use $P_{11}, P_{12}, P_{23} - d, P_{23} - e$, complement the first two, and take anchors a_1, b_3 and privates b_2, b_1, e, d.

The degree hypotheses and the directional degree bound now give $d^+(U_3) = d^+(P_{23}) = 3$. The outgoing labels of U_3 must include both b_1, b_2. Otherwise two labels c, e lie outside $A \cup \{b_1, b_2, b_3\}$; then $U_1, U_2, U_3 + c, U_3 + e$ have two anchors in A and privates b_1, b_2, c, e. Similarly, the two successors of P_{23} other than U_3 must include $P_{23} + b_2$. If their labels c, e both differ from b_2, the sets $L_1, P_{22}, P_{23} + c, P_{23} + e$ have two anchors in $A \setminus \{a_1, a_2\}$ and privates a_2, b_2, c, e.

Finally the cube vertices are

$$L_2, A, P_{22}, U_2, P_{23}, U_3, P_{23} + b_2, U_3 + b_2.$$

All anchor choices above exist by balance. □

7 Sunflowers and the quadratic small-side boundary

Lemma 7.1 (Elementary sunflower bound)**.** *A family of more than $j!(q-1)^j$ distinct j-sets contains a q-sunflower, for $q \ge 2$ and $j \ge 1$.*

Proof. For $j = 1$, any q distinct singletons work. Induct on j. If q pairwise disjoint sets exist, they form a sunflower with empty core. Otherwise a maximal pairwise disjoint subfamily has at most $q - 1$ members, and its union W meets every member of the family. Since $|W| \le j(q-1)$, some $x \in W$ lies in more than $(j-1)!(q-1)^{j-1}$ members. Delete x from these members and use induction. Restoring x adds it to the core. □

7.1 A fixed-rank boundary can be widened at quadratic cost

Lemma 7.2. *For every fixed integer $k \geq 2$, the number of splits with a side of size at most k is at most*

$$n + 1 + \binom{n}{2} \sum_{j=0}^{k-2} j!3^j.$$

In particular this number and the cost of deleting all such splits and their incident edges are $O_k(n^2)$.

Proof. Choose one smaller side of every nontrivial split under consideration. Fix two taxa h_1, h_2, and consider the chosen sides containing both. After removing this pair, at most $k-2$ taxa remain in each side. In any one cardinality layer j, more than $j!3^j$ distinct remainders contain a four-sunflower, by the usual sunflower lemma. Its petals are nonempty. The two fixed taxa are common anchors, and one point from each petal is a private taxon of that side. This is a forbidden configuration. Thus the codegree of every pair is at most $\sum_{j=0}^{k-2} j!3^j$. Each chosen nontrivial side contains at least one pair; summing pair codegrees gives the bound, and at most $n+1$ trivial or empty splits remain. A nontrivial vertex has degree at most six. The trivial and empty splits have total incident degree $O(n^2)$, proving the edge claim. This lets later local arguments use any fixed wider balance threshold without changing either original asymptotic target. □

8 Structure of second overlaps: large links are paths

We classify the large second links by a parity restriction and a sunflower count.

Let $\mathcal{G}$ be a split system on an old ground set Y whose full two-lift is 2-weakly compatible. Orient its sides away from an old root. An immediate necessary condition is that $\mathcal{G}$ avoids all four singleton splits on four taxa: orient such traces as singletons and use the two new taxa as common anchors in the full lift.

8.1 A three-column parity restriction

Lemma 8.1. *For three distinct base splits, record the eight old membership types in $\{0,1\}^3$. If at least three types of one parity occur, each of those occurring types has multiplicity one.*

Proof. Otherwise choose a repeated type p and two other occurring types u, v of the same parity. Their differences from p flip two coordinates each, with one common flipped coordinate r. Name the other coordinates s, t so that $u = p \oplus e_r \oplus e_s$ and $v = p \oplus e_r \oplus e_t$. Use four lifted columns, taking base column r twice and base columns s, t once each. The repeated old taxa give anchor pattern $p' = (p_r, p_r, p_s, p_t)$. The old types v, u give the third and fourth private patterns $p' \oplus (1111 \oplus e_3)$ and $p' \oplus (1111 \oplus e_4)$. Assign the first two required private patterns to the two free taxa. Complement columns to turn p' into 1111. This is the forbidden configuration. The two repeated-base columns are distinguished by the free patterns, so all four selected lifted splits are distinct. The argument is invariant under changing the orientations of any base columns. □

8.2 A crossing pair constrains all other sides

Two splits cross if all four intersections of their sides are nonempty. Suppose A, B are crossing rooted sides in $\mathcal{G}$. For any third rooted side C, either C is a Boolean function of membership in A, B, or it differs in at most four old taxa from $A \triangle B$ or its complement.

Indeed, let the two parity classes of the eight (A, B, C) types be considered. If both have at most two occurring types, at most four types occur in total. All four (A, B) cells are nonempty,

so each cell has exactly one C value. This makes C a Boolean function of A, B. If one parity has at least three types, Lemma 8.1 puts at most four old taxa in that parity. Flipping C on just those taxa makes C equal to one of the two parity functions of A, B.

We need a bound for the two Hamming balls in this covering statement. For a fixed template B_0 and an integer k, a family avoiding four singleton traces has at most

$$b_k = \sum_{j=0}^{k} j!4^j \tag{8.1}$$

rooted sides at set distance at most k from B_0. No balance assumption on the template is needed.

To prove this, more than $j!4^j$ distinct difference sets of size j contain a five-sunflower, by Lemma 7.1. Its petals are nonempty. Call a petal positive if it contains a taxon outside B_0, and negative otherwise. Choose four petals having an even number of negatives; this is always possible among five. Pick a private point outside B_0 from a positive petal and inside B_0 from a negative petal. Their four membership rows are e_i for positive petals and $1111 \oplus e_i$ for negative petals. With zero negatives they already form I_4; with four negatives complement all columns. With two negatives, complement the two positive columns: the private rows become the four unit vectors, with the two columns within each sign group interchanged. Thus every case gives four singleton split traces, a contradiction. Sum over distance layers to obtain (8.1).

Theorem 8.2. *If $\mathcal{G}$ contains a crossing pair, then*

$$|\mathcal{G}| \le 14 + 2b_4 = 13144.$$

The constant is conservative and is not claimed optimal.

Proof. The crossing pair gives sixteen exact Boolean templates and two radius-four balls about its symmetric difference and complement. The latter templates are already two of the sixteen. Thus at most fourteen sides lie outside the two balls. Each ball contains at most $b_4 = 1 + 4 + 32 + 384 + 6144 = 6565$ sides, as just proved. □

8.3 Compatible large links have no branching

Theorem 8.3. *If $|\mathcal{G}| > 13144$, then $\mathcal{G}$ is a path split system: its nonempty splits can be represented by the cuts of one path with the old taxa assigned to vertices. In particular, for every original 2-weakly compatible system,*

$$|\partial_x\partial_y\mathcal{S}| > 13144 \quad \Longrightarrow \quad \partial_x\partial_y\mathcal{S} \text{ is a path split system.}$$

Proof. Theorem 8.2 makes $\mathcal{G}$ compatible. Use its tree representation, contracting edges whose splits are absent. Every leaf carries old taxa; an unlabeled degree-two vertex would give two identical edge splits, so can be suppressed. If a branching vertex has degree at least four, pick one taxon in each of four incident components. The four edge splits isolate those taxa, contrary to the four-singleton prohibition. At a degree-three vertex, orient its three incident edge sides as the three disjoint components. Their old row patterns include the three unit vectors. By Lemma 8.1, every component contains exactly one taxon. It consequently has only one edge in the reduced tree: further branching would need more leaves and an unlabeled subdivision would duplicate a split. Thus the whole tree is a three-star, with at most three nonempty splits and the possible empty split. This contradicts the assumed size. No branching remains, so the tree is a path.

The original second overlap is full-two-liftable as a subsystem of $\mathcal{S}$, which proves the final assertion. □

This classifies sufficiently large pair links for the full original problem. It does not itself count how many such links can coexist.

8.4 Deleting non-path links at quadratic cost

Write $K = 13144$. A path split system means a subfamily of the edge cuts of one taxon-labeled path, together with the possible empty split. It need not be a connected Hamming path: selected cuts may skip edges, and several taxa may share a path vertex.

Theorem 8.4. *Every 2-weakly compatible system $\mathcal{S}$ on n taxa has a subsystem $\mathcal{S}'$ obtained by deleting at most $K\binom{n}{2}$ splits such that every second derivative of $\mathcal{S}'$ is a path split system or is empty.*

Proof. For each unordered pair $\{x, y\}$, let $G_{xy} = \partial_x \partial_y \mathcal{S}$. If G_{xy} is not a path system, the large-link theorem above gives $|G_{xy}| \leq K$. Each split in G_{xy} corresponds to all four placements of x, y in $\mathcal{S}$. Choose a nontrivial corner for deletion. Such a corner always exists for $n \geq 4$: if both old sides are nonempty, place x, y on different sides; if one old side is empty, place both new taxa there. The case $n \leq 3$ is vacuous, as there are no non-path second links. Let D be the union of all these choices over the non-path pairs. Then

$$|D| \leq \sum_{\{x,y\}:G_{xy} \text{ not a path}} |G_{xy}| \leq K\binom{n}{2}.$$

Set $\mathcal{S}' = \mathcal{S} \setminus D$. Every full square for an originally non-path pair has lost a corner, so its new second derivative is empty. For an originally path pair, the new derivative is a subfamily of G_{xy}, and hence remains a path system. Removing splits preserves 2-weak compatibility. This proves the assertion. □

9 Normalization of degree-six neighborhoods

We reduce an arbitrary counterexample to local charge proximity to a finite domain.

9.1 Every local change uses an initial neighbor label

Let A be nontrivial and have degree six. Write its three incoming labels as $I = \{a_0, a_1, a_2\}$ and its three outgoing labels as $O = \{b_0, b_1, b_2\}$.

Lemma 9.1. *Let B be another member. Then*

$$\begin{aligned} |X \setminus (A \cup B)| \geq 2, \quad A \setminus B \neq \varnothing \quad &\Longrightarrow \quad (A \setminus B) \cap I \neq \varnothing, \\ |A \cap B| \geq 2, \quad B \setminus A \neq \varnothing \quad &\Longrightarrow \quad (B \setminus A) \cap O \neq \varnothing. \end{aligned}$$

Proof. For the first implication, suppose B retains all three a_i but omits some $c \in A \setminus I$. Take the four columns

$$A - a_0, \quad A - a_1, \quad A - a_2, \quad B.$$

The rows a_0, a_1, a_2, c have the four complement-of-singleton patterns. Any two taxa outside $A \cup B$ are zero anchors, giving the forbidden configuration. For the second implication, suppose B contains no b_i but contains some $d \in B \setminus A$. On the four columns

$$A + b_0, \quad A + b_1, \quad A + b_2, \quad B,$$

the rows b_0, b_1, b_2, d are the four singleton patterns. Two taxa in $A \cap B$ are all-one anchors. Again this is forbidden. □

Corollary 9.2. *If $|A|, |X \setminus A| \geq 4$, graph distance from A agrees with Hamming distance on its closed graph ball of radius four.*

Proof. A member at Hamming distance two must change at least one of the six initial neighbor labels, by Lemma 9.1. The needed anchor intersections have size at least two because there are only two changes. Flipping that initial label first gives a graph path of length two to the member. Thus every present Hamming-distance-two member is at graph distance two.

Graph and Hamming distances have the same parity. A vertex first reached at graph distance three can only have Hamming distance three or one; distance one would already be an edge. A vertex first reached at graph distance four can only have Hamming distance four, two, or zero. The latter two cases have already been reached earlier. This proves the assertion. □

This point matters for finite models. In an arbitrary induced cube subgraph, graph distance four can occur at Hamming distance two. The present split-system lemma excludes those detours; one cannot omit them merely from a graph-radius argument.

9.2 Two further restrictions at each protected six-vertex

Lemma 9.3. *Work in a retained family with two constant inside taxa and two constant outside taxa. Let D have degree six and suppose all its neighbors are retained. If $L = D - a$ and $L + b$ are present, with $b \notin D$, then $D + b$ is present. The reversed statement holds for an upper neighbor and an incoming two-edge path.*

Moreover, for each lower neighbor $L = D - a$, at most one of its incoming labels is absent from the incoming-label triple of D. For each upper neighbor, at most one outgoing label is absent from the outgoing-label triple of D.

Proof. If b is not one of the three outgoing labels of D, the three successors of D, together with $L + b$, have four singleton private rows. Two taxa in L are all-one anchors. Thus b must be an outgoing label of D, giving $D + b$. Reversal gives the incoming statement.

For the second claim, suppose $L = D - a_i$ has two predecessors $L - c_1, L - c_2$ with c_1, c_2 outside the initial incoming triple. Let a_j, a_k be the other two incoming labels of D. The four columns

$$L - c_1, \quad L - c_2, \quad D - a_j, \quad D - a_k$$

have complement-of-singleton rows c_1, c_2, a_j, a_k. Two outside anchors give a forbidden configuration. The outgoing claim follows by reversal. □

These implications apply to every retained six-vertex whose immediate neighbors are preserved, not just to the original center. They are necessary compatibility conditions; they do not assert that the whole family is closed under arbitrary square completions.

9.3 An ambient-independent 2403-member domain

Assume now $|A|, |X \setminus A| \geq 16$, and retain the closed graph ball of radius four. At each first-appearance level $r = 1, 2, 3, 4$, there are at most three new incoming labels and three new outgoing labels. Here is the induction explicitly. If four new incoming labels first appear at level r, choose one level-r member omitting each of them. Its last edge changes that new label, so it still contains the other three. All outgoing labels changed in these four members appeared before level r. Their union has size at most $3(r - 1)$. Consequently at least $16 - 3(r - 1) \geq 7$ outside taxa remain as common zero anchors. Four private rows give a contradiction. The outgoing case is symmetric, using common inside anchors.

Thus at most twelve labels of each sign vary. Choose two constant taxa of each sign, including the root on the absent side, and delete the other constant taxa. At least four were available on either side. The retained family has at most 28 taxa, with all pairwise Hamming differences preserved. Degrees through graph radius three are preserved, since all their neighbors lie in the retained radius-four ball.

Use the following fixed prototype. The potential incoming labels at levels one through four are 0–2, 3–5, 6–8, 9–11. The outgoing groups are 14–16, 17–19, 20–22, 23–25. Taxa $12, 13$ are constant inside and $26, 27$ constant outside, with root 27. The base mask is $A_0 = 16383$. Unused potential labels are inactive; inactive incoming slots are constant dummy bits and cannot serve as witness taxa.

For a flip set S, put

$$I_* = \{0, \dots, 11\}, \quad O_* = \{14, \dots, 25\}, \quad I_1 = \{0, 1, 2\}, \quad O_1 = \{14, 15, 16\}.$$

Lemma 9.1 gives the two conditions

$$S \cap I_* \neq \varnothing \implies S \cap I_1 \neq \varnothing, \qquad S \cap O_* \neq \varnothing \implies S \cap O_1 \neq \varnothing.$$

All needed anchors exist for these radius-four members. Starting with the empty flip set, at step r add one unused label from the first r groups of either sign and retain these two conditions. By Corollary 9.2, every source member in the radius-four ball occurs in this construction. The successive numbers are

$$1, \quad 6, \quad 33, \quad 236, \quad 2127, \qquad \text{total } 2403.$$

The counts are a domain size, not the number of members of a counterexample. A selected member at level r must still have a selected parent at level $r - 1$, and the optional-label flags must match their first appearances.

The independent script `verify_radius4_domain.py` first reconstructs the larger 6096-member domain including all 51 possible two-flip detours. It then checks a direct six-taxon forbidden witness for each of the 3693 excluded members. There are no extra exclusions requiring a further parent argument. It reconstructs the 2403-member domain separately by sequential paths. The check passed, and its certificates are in `radius4_domain_verification.json`.

9.4 What path second links contribute to the model

This particular normalization preserves compatibility of second links if the original family has path second links. Deleting splits introduces no new full faces. Deleting only constant rows of the retained family neither merges members nor creates a new full face on varying coordinates. Restricting the old second link to the remaining taxa preserves its compatibility. The general counterexample to preservation under arbitrary taxon deletion does not apply to this constant-row deletion.

Every retained full face on varying coordinates has both constant inside taxa in its base and both constant outside taxa outside its base. Therefore any two bases B, C on the same coordinate pair have nonempty intersection and nonempty common outside. If the two link splits are compatible, one of $B \setminus C$ and $C \setminus B$ must be empty. Hence

$$B \subseteq C \quad \text{or} \quad C \subseteq B.$$

The bases on each coordinate pair form an inclusion chain. In the anchored prototype there are 2343 possible full faces and 12726 pairs of incomparable bases. A source counterexample from the path-pruned class cannot select both faces in any of those pairs.

9.5 First appearances of labels in radius-three balls

The following version also applies when the center has degree five. Suppose both sides of a center A have size at least eleven and retain its entire closed graph ball of radius three. Classify a taxon by the first level at which a path from A changes it, and by its initial membership in A. At each of levels one, two, and three, at most three labels of either sign first appear.

Indeed, if four outgoing labels first appear at level ℓ, choose a level-ℓ vertex first using each one. Its predecessor uses only earlier labels, so the four labels are private to their four chosen columns. At most $3(\ell-1)$ earlier incoming labels have appeared. At least $11-3(\ell-1) \geq 5$ taxa of A are therefore common anchors. This is forbidden. The incoming argument uses outside anchors and complements the four columns. This proves the assertion by induction on the level.

Thus at most nine taxa of either sign vary. Keep two constant taxa of each sign, including a root outside, and delete the others. All differences between retained members and all degrees through radius two are preserved. Compatibility is preserved by restriction. For a degree-five center with two incoming and three outgoing labels, the two sign totals improve to eight and nine, respectively, so the normalized ground set has at most $8+9+4=21$ taxa. For a degree-six center the bound is 22. Unused optional slots cannot be used as physical witness taxa.

10 Uniform regularity of cubes in high-degree neighborhoods

The result is computer-assisted: a hand normalization covers arbitrary ambient size, and separately written finite encodings both verify the resulting unsatisfiable cases.

Fix the rooted Hamming graph. A cube with directions $D=\{x,y,z\}$ and base A, disjoint from D, consists of all $A \cup U$ with $U \subseteq D$. Call it balanced if

$$|A| \geq 5, \qquad |X \setminus (A \cup D)| \geq 5.$$

Its closed neighborhood comprises the cube and every present vertex joined to a cube vertex by one Hamming edge.

Theorem 10.1. *Let a balanced cube have no vertex of degree at most four in its closed neighborhood. Then every cube vertex has degree exactly five, and there is a unique $s \in D$ such that, at every cube vertex B,*

$$d^+(B) = 3 - \mathbf{1}_{s \in B}, \qquad d^-(B) = 2 + \mathbf{1}_{s \in B}.$$

Furthermore, every present vertex in the cube's closed neighborhood has its s-flipped partner in the same closed neighborhood.

A balanced cube satisfying the closed-neighborhood hypothesis is called a *good cube*. The neighboring vertices may have degree five or six.

10.1 Bounding the number of local labels

Let $\mathcal{L}_0$ be the cube and its closed neighborhood, and retain $\mathcal{L}_1$, consisting of $\mathcal{L}_0$ and all its original graph neighbors. Removing other vertices leaves every degree at $\mathcal{L}_0$ unchanged and preserves compatibility. The retained vertices differ from some cube vertex in at most two coordinates outside D.

Let $I_0 \subseteq A$ be the labels removed on edges from the cube and let $O_0 \subseteq X \setminus (A \cup D)$ be the labels added there. Then

$$|I_0| \leq 3, \qquad |O_0| \leq 3. \tag{10.1}$$

For four different outgoing labels, choose one cube neighbor for each. Those labels are private and any two taxa in A are anchors. Four incoming labels give the complementary obstruction, with two anchors outside $A \cup D$.

Let I_1, O_1 be additional incoming and outgoing labels used in the second neighborhood. Every vertex using a new label flips that label and one initial label: it is a neighbor of a present first-neighborhood vertex. Thus a column using a new outgoing label contains no other new outgoing label, and every such column contains $A \setminus I_0$. Four different new outgoing labels

would consequently be private, with two anchors in $A \setminus I_0$, whose size is at least two. Similarly, complement four columns with new incoming labels and use two anchors outside $A \cup D \cup O_0$. Hence

$$|I_1| \leq 3, \qquad |O_1| \leq 3. \tag{10.2}$$

Put $I = I_0 \cup I_1$ and $O = O_0 \cup O_1$. There are at least three used labels of each sign. At the bottom cube vertex there are at least two incoming edges, since its three cube edges point outwards and its degree is at least five. An incoming neighbor has degree at least five and outdegree at most three, so it has at least two incoming edges. Their labels, together with the removed label, give $|I| \geq 3$. At the top cube vertex the reversed argument gives $|O| \geq 3$.

All remaining old taxa have constant membership on $\mathcal{L}_1$. Retain at most two constantly present and two constantly absent taxa, including the old root among the absent taxa. This preserves the obstruction criterion: constant row multiplicities need only be known up to two. It also preserves every Hamming distance between retained members, so degrees and their directions are unchanged.

Balance is preserved. For example, the normalized base size is

$$|I| + \min\{|A \setminus I|, 2\} \geq 5,$$

because $|I| \geq 3$ and the original $|A| \geq 5$. The outside bound is the same. The normalized ambient size is therefore at most

$$3 + 3 + 3 + 3 + 3 + 2 + 2 = 19. \tag{10.3}$$

This is a uniform counterexample reduction, rather than an assumption that the original ambient size is small.

10.2 The finite model and its three exclusions

Use the following conceptual coordinates, making unused coordinates optional:

I_0	$0, 1, 2$
I_1	$3, 4, 5$
constant inside	$6, 7$
D	$8, 9, 10$
O_0	$11, 12, 13$
O_1	$14, 15, 16$
constant outside	$17, 18$ (18 is the root).

The conceptual base is 255. For each of its eight cube placements, allow no external flip, one initial flip, or two external flips with at least one initial label. This gives exactly

$$8\left(1 + 6 + \binom{12}{2} - \binom{6}{2}\right) = 464$$

possible vertices.

Every selected vertex flipping a label requires that taxon to be active. An initial label is active exactly when it occurs on an edge from the cube; a new label is active exactly when it is used at the second level. Every selected second-level vertex must adjoin a selected first-level vertex. Constant-coordinate flags allow zero, one, or two inside taxa and one or two outside taxa, always retaining the root. Prefix symmetry within each label group is valid by relabeling. Permuting cube directions allows the three first-level outdegrees to be sorted.

All present vertices have both directed degrees at most three; every present cube vertex and first neighbor has degree at least five. The implementation's optional constant-row guards

are equivalent to requiring normalized balance, as checked separately: the degree constraints already force at least three used labels of each sign.

Compatibility is imposed by necessary six-taxon trace prohibitions. For two enabled anchors and four enabled private taxa, the conjunction of the four corresponding restricted splits is forbidden. Each such cut is valid for any compatible system, regardless of the other enabled taxa. It is enough for these necessary cuts to make the model unsatisfiable; equivalence to the full compatibility condition is not required for this direction.

Three finite exclusions prove Theorem 10.1:

1. Allow cube degrees five or six and neighboring degrees at least five, but require a degree-six corner. This model is unsatisfiable using 3011 certified trace cuts.

2. Require all cube degrees to be five, allow neighboring degrees at least five, and prohibit all three possible fixed-axis outdegree patterns. This model is unsatisfiable using 2726 certified trace cuts.

3. Require one fixed-axis pattern, but require a first-neighborhood vertex to lack its axis-flipped partner. This model is unsatisfiable using the same 3011 certified trace cuts as the first exclusion.

Each real counterexample maps to one of these models by the preceding normalization. The first exclusion makes all cube corners degree five; the second then supplies the axis. It is unique because the three axis patterns differ on the cube.

`research_cube_phase_uniform.py` produces the cases and records their trace witnesses. The separate `verify_cube_phase_uniform.py` enumerates the 464 vertices independently, reconstructs every six-taxon witness, expresses degrees using arithmetic sums instead of the producer's pseudo-Boolean rules, and replays the unsatisfiability checks. Both encodings use Z3. The successful replay records are `cube_phase_uniform_high_verification.json` and `cube_high_core_six_verification.json`, together with `cube_unpaired_axis_verification.json` for the stronger pairing claim. They also verify that the optional-row balance guards cannot admit a smaller balanced side. The precise proof status is computer-assisted, with an ambient-independent hand normalization and separately encoded finite replays.

11 The forced grid cross around a good cube

The axis theorem gives the following precise neighborhood after two further finite exclusions.

Theorem 11.1. *Let a balanced cube be good, with axis s and other directions x, y. It uses exactly two incoming external labels a, b and exactly two outgoing external labels c, d. These four labels and s, x, y are distinct. After naming them appropriately, put $K = A \setminus \{a, b\}$ and*

$$P_i = \{a, x, c\}_{\le i}, \qquad Q_j = \{b, y, d\}_{\le j}, \quad 0 \le i, j \le 3,$$

where the notation means the prefix of the displayed ordered list. The cube's entire closed neighborhood is exactly

$$\{K \cup P_i \cup Q_j \cup U : U \subseteq \{s\},\ (i, j) \in R\},$$

where

$$R = \{1, 2\}^2 \ \cup (\{0, 3\} \times \{1, 2\}) \ \cup (\{1, 2\} \times \{0, 3\}).$$

Thus it has exactly 24 vertices: a rectangular cross times an axis edge.

Proof. The nineteen-taxon normalization and finite model in Section 10 remain applicable. Require a valid fixed-axis pattern and three active initial incoming labels. This is unsatisfiable

using the same 3011 six-taxon cuts as the degree-six exclusion. The outgoing counterpart is separately unsatisfiable. Both were replayed with the separate arithmetic encoding. The data are `cube_three_initial_verification.json` and `cube_three_outgoing_verification.json`; both replays passed. The normalization covers all possible counterexamples, so at most two labels of each sign occur on edges from a good cube. At its bottom corner there are two incoming edges, and at its top corner two outgoing edges. Therefore both label counts are exactly two.

For the remainder the proof is by hand. Work on the side where s is absent. Its four square corners have transverse indegrees and outdegrees two each, after the axis edge is removed. Thus at $A + x$ and $A + y$ exactly one incoming external label and one outgoing external label occur. If both incoming labels were a, then $A - a$ would have the four outgoing neighbors

$$A, \quad A - a + x, \quad A - a + y, \quad A - a + s.$$

The axis-pairing theorem supplies the last one. This contradicts the directed degree bound. Name the labels so that $A + x$ removes b and $A + y$ removes a.

If both outgoing labels at these two corners were c, then $A + x + y + s + c$ would have four incoming neighbors, obtained by removing c, y, x, s, respectively. The three first-neighborhood members involving c have their axis partners, so all four neighbors exist. This is again impossible. Name the outgoing labels so that $A + x$ adds c and $A + y$ adds d. The bottom removes a, b and the top adds c, d. These are precisely the eight external neighbors described by R on the absent-axis side. Pairing gives their eight partners. Together with the eight cube vertices these are the entire neighborhood, since every cube corner has exactly two external neighbors and degree five. □

12 The entire good-cube part has quadratic size

A finite foreign-face classification bounds each transverse second link, which permits iterative deletion at quadratic cost.

Theorem 12.1. *Let C be a good balanced cube with axis s, other directions x, y, and bottom A. Then*

$$\partial_x \partial_y \mathcal{S} \subseteq \{\varnothing, A, A + s\}.$$

The last two members always occur. The empty member may occur. Compatibility of this second overlap is not an additional hypothesis.

Proof. Use Theorem 11.1. Write the two transverse chains as $(a, x, c), (b, y, d)$, and put $K = A \setminus \{a, b\}$. Let L be the taxa outside $K \cup \{a, x, c, b, y, d, s\}$. Balance gives

$$|K| \geq 3, \qquad |L| \geq 3.$$

The root lies in L. For an arbitrary old side $B \in \partial_x \partial_y \mathcal{S}$, all four placements of x, y occur. Together with the 24-vertex cross they form a compatible subfamily.

Apart from x, y, the seven varying cross labels leave five freely chosen old membership bits, namely those of a, c, b, d, s. There are 32 possibilities. Partition the constant taxa into

$$K \cap B, \quad K \setminus B, \quad L \cap B, \quad L \setminus B,$$

and cap their four multiplicities at two. This cap preserves the forbidden-configuration criterion: row types need only be known as absent, present once, or present at least twice. In particular it preserves all possible private patterns and repeated anchor patterns.

For the two K categories, the five possible saturated pairs are

$$(0, 2), (2, 0), (1, 2), (2, 1), (2, 2).$$

The pair $(1,1)$ is excluded by $|K| \geq 3$. For the L categories, $L \setminus B$ contains the root. Their four possible pairs are

$$(0,2),(1,2),(2,1),(2,2).$$

Thus there are exactly $32 \cdot 5 \cdot 4 = 640$ cases. All are covered by the following finite classification:

(a,c,b,d,s) in B	$K \cap B$	$K \setminus B$	$(L \cap B, L \setminus B)$
$(0,0,0,0,0)$	0	2	$(0,2)$
$(1,0,1,0,0)$	2	0	$(0,2)$
$(1,0,1,0,1)$	2	0	$(0,2)$.

Every other case has an explicit four-member, six-taxon forbidden witness. The 637 witnesses are stored in `cube_foreign_face_classification.json`. `verify_cube_foreign_face.py` checks each witness, separately reconstructs all 640 cases and the cross, and verifies exact coverage. The successful record is `cube_foreign_face_verification.json`. The producer uses NumPy; the verifier uses neither NumPy nor a solver.

For completeness, each saturated case has a representative satisfying $|K|, |L| \geq 3$. If a sole nonzero category has saturated size two, retain three representatives there; otherwise its saturated counts already total at least three. This never changes any obstruction and can be realized by restriction of the original categories. Retain the root in $L \setminus B$. Consequently the finite classification covers arbitrary ambient size, not merely small examples.

The three accepted cases respectively say that B is empty, equals $K \cup \{a,b\} = A$, or equals $A + s$. The last two full faces are the original cube, proving the assertion. □

Corollary 12.2. *On n taxa with a fixed root, there are at most*

$$\binom{n-1}{2}$$

good balanced cubes. The union of all their closed neighborhoods has at most $24\binom{n-1}{2}$ vertices.

Proof. Each good cube has a unique axis and hence a unique transverse pair x, y, neither equal to the root. Fix such a pair. If one good cube occurs, its second overlap is contained in $\{\varnothing, A, A+s\}$. Since $|A| \geq 5$, the only Hamming edge among these three sides is $A, A+s$. Any other good cube assigned to the same transverse pair would require another edge, or the same edge with the same eight vertices. Thus there is at most one cube per pair. Its closed neighborhood has exactly 24 vertices, so a union bound gives the second assertion. □

12.1 An iterative reduction that removes every good cube

Theorem 12.3. *Deleting at most $8\binom{n-1}{2}$ splits leaves a rooted family with no balanced good cube. Every deleted split is nontrivial. If path-link pruning is performed first, the final family also has every second overlap a path system, with total deletion at most*

$$13144\binom{n}{2} + 8\binom{n-1}{2}.$$

An edge inequality with coefficient strictly below $5/2$ transfers across this reduction with $O(n^2)$ additive loss.

Proof. As long as a balanced good cube remains, remove its eight corners. Charge this step to its unique transverse pair x, y. At that moment Theorem 12.1 says the pair link is contained in $\{\varnothing, A, A+s\}$. Removing all eight corners destroys both full faces $A, A+s$, leaving at most the empty old side in the link. Links can only lose members when original splits are deleted. Thus

the same pair can never support another cube, and in particular can never be charged again. There are at most $\binom{n-1}{2}$ steps. This argument accommodates good cubes created by previous deletions; it does not assume a single simultaneous removal suffices.

Each removed corner has both sides of size at least five, hence is nontrivial. Path-link pruning deletes only nontrivial splits too, and all further deletions preserve its path property. If D splits are removed, their original degrees are at most six, so the edge loss is at most $6D$. An inequality $E' \le \alpha V' + Cn^2$ for the final class transfers as

$$E \le \alpha V + Cn^2 + (6 - \alpha)D.$$

For fixed $\alpha < 5/2$ this is an $O(n^2)$ additive loss. □

13 The degree-five bulk estimate

The following computer-assisted completion theorem controls regular degree-five regions.

Theorem 13.1. *Suppose $|A|, |X \setminus A| \ge 11$ and every vertex in the closed graph ball of radius two about A has degree exactly five. Then a full three-dimensional cube contains A.*

Proof. Retain the graph ball of radius three, preserving degrees through radius two. The first-appearance argument from Section 9.5 bounds each subsequent level by three incoming and three outgoing labels. Reverse this retained local family if necessary to arrange two initial incoming and three initial outgoing labels. There are at most eight varying incoming and nine varying outgoing labels. Two constant taxa of each sign remain available; retain them, including a root on the absent side. This reduces every possible counterexample to at most 21 taxa, preserving compatibility and the tested degrees. The domain is

$$1 + 5 + \left(\binom{11}{2} - \binom{6}{2}\right) + \left(\binom{11}{3} - \binom{6}{3}\right) + 6\left(\binom{11}{2} - \binom{6}{2}\right) = 431$$

possible members. Optional first-appearance flags and parent constraints have the same meaning as in the degree-six model. All retained members are nontrivial because the four constant taxa are retained. A new distance-three member must change three labels; changing just one would give an additional immediate neighbor of the degree-five center.

Write the two incoming labels as a_1, a_2 and the three outgoing labels as b_1, b_2, b_3. A swap graph edge (i, j) means that $A - a_i + b_j$ occurs. Each incoming neighbor $L_i = A - a_i$ has outdegree $1 + \deg(i)$: an additional outgoing label outside the b-triple, together with the three $A + b_j$, would give four private taxa and two anchors in $A - a_i$. Its degree five forces the swap row degree to be one or two.

A rectangle is impossible because the rectangle theorem supplies a corner of degree at most four within distance two. Four remaining swap edges would form the five-vertex path treated in Proposition 6.1. Its endpoints have degree five, so a cube already contains A. Thus a counterexample has at most three swap edges. Up to row and column permutations the possibilities are

shape	edges	number of labeled graphs
matching	$(1,1),(2,2)$	6
common column	$(1,1),(2,1)$	3
three-edge path	$(1,1),(1,2),(2,2)$	12
disjoint star and edge	$(1,1),(1,2),(2,3)$	6.

Here are the further hand-implied constraints used in the finite models. Every shape has a row of degree one, whose first neighbor L_i has outdegree two and indegree three. The other

initial incoming label a_{3-i} must be among these three incoming labels. Otherwise the three predecessors of L_i together with $A - a_{3-i}$ have four complementary private patterns and two common outside anchors. Consequently $A - a_1 - a_2$ occurs, and at least two new incoming labels first appear at level two. Prefix symmetry permits the first two optional labels in that group to be active.

An outgoing neighbor $U_j = A + b_j$ with an empty swap column has indegree exactly two. Indegree three would give two additional predecessors deleting labels outside $\{a_1, a_2\}$. Together with $A - a_1, A - a_2$ these provide four complementary private patterns and two outside anchors. Degree five then gives outdegree three. A column of degree two instead forces indegree three. These constraints are necessary, rather than extra assumptions about the counterexample.

For each of the four shapes impose degree five through radius two, directed degree at most three, the preceding necessary constraints, and the absence of a cube through A. Only triples of the center's five initial directions need be tested. Each of these four finite models is unsatisfiable with the same 6571 necessary six-taxon cuts.

`research_five_regular_ball.py` produces the models and `verify_five_regular_ball.py` separately enumerates the 431 vertices, checks all four shape classes, audits 1450 explicit seed witnesses, reconstructs the degrees and cube constraints arithmetically, and replays each exclusion. The four successful records are:

Matching	`five_regular_ball_matching_verification.json`
Common column	`five_regular_ball_common_verification.json`
Three-edge path	`five_regular_ball_path_verification.json`
Disjoint star and edge	`five_regular_ball_disjoint_verification.json`.

Both encodings use Z3. The proof is computer-assisted with separately written finite replays, not an independently checked formal solver certificate. The normalization and four-case coverage complete the uniform proof. □

13.1 Where a degree-five vertex can be charged

Corollary 13.2. *In a family with no balanced good cube, every degree-five vertex A with both sides of size at least eleven has either a positive-charge vertex within graph distance four, or a degree-six vertex within graph distance six.*

Proof. Suppose neither exists. Every vertex within distance four is balanced, so it has zero charge. A degree-at-most-three vertex retains positive charge. A degree-four zero vertex must send its unit to a degree-six recipient or owner at distance at most two. This would give a degree-six vertex within distance six of A. Such vertices are excluded, so the entire radius-four ball has degree five. The theorem gives a cube through A, whose corners have distance at most three from A and whose closed neighborhood has distance at most four. All its neighbors have degree five. Its base and outside complement have size at least eight, so it is a balanced good cube, a contradiction. □

For the application, all members have both sides of size at least sixteen. Let p count vertices with $C_2 \geq 1$ and H count degree-six vertices. Maximum degree six gives the graph-ball bounds

$$B_r = 1 + 6\sum_{j=0}^{r-1} 5^j, \qquad B_4 = 937, \quad B_6 = 23437.$$

Proposition 13.3. *If all members have both sides of size at least sixteen and there is no balanced good cube, then*

$$|V| \leq 938p + 23447H.$$

Proof. By Corollary 13.2, assign each degree-five vertex to a positive-charge vertex at distance at most four or a degree-six vertex at distance at most six. There are at most $937p + 23437H$ such vertices. A vertex of degree at most three retains positive charge: a degree-two stage-one donor can lose two of its three units, other stage-one donors lose at most one, and a stage-two donor makes no first-stage donation and loses at most one. A zero-charge degree-four vertex sends its unit to a degree-six recipient or owner. Each such recipient has at most six adjacent donors and three partner donors. Thus there are at most $9H$ zero-charge degree-four vertices. Positive vertices of degrees at most four contribute at most p, and degree-six vertices contribute H. Adding yields the claim. □

14 Degree-six squares have quadratic total size

Full squares with degree-six corners can be removed at quadratic cost. The local classification below is computer-assisted, but uses explicit forbidden witnesses rather than a solver for its verification.

Definition 14.1. For distinct nonroot taxa $x, y \notin A$, write

$$S_{xy}(A) = \{A, A + x, A + y, A + x + y\}.$$

A *six-square* is such a full square whose four vertices have degree six in the original split graph. Its base is A.

Lemma 14.2. *Suppose $S_{xy}(A)$ is a six-square, with $|A| \geq 5$ and $|X \setminus A| \geq 7$. There are taxa $t \in A$ and $b \notin A \cup \{x, y\}$ for which both $S_{xy}(A - t)$ and $S_{xy}(A + b)$ are full squares.*

Proof. Each corner has three incoming and three outgoing neighbors. Call an incoming label external if it is not x or y, and similarly for an outgoing label. All external incoming labels lie in A. Their union has size at most three: four distinct labels would give four predecessors, one removing each label. On these four columns the labels have the four complement-of-singleton patterns. Any two taxa outside $A \cup \{x, y\}$ are anchors with pattern zero, producing the forbidden configuration. Likewise, four distinct external outgoing labels give four successors with singleton patterns, anchored by two taxa in A.

The bottom corner already has three external incoming labels, and the top already has three external outgoing labels. Thus both unions have size exactly three. Name them a_0, a_1, a_2 and b_0, b_1, b_2. At the middle corners the incoming labels are two-element subsets of the a-triple, and their outgoing labels are two-element subsets of the b-triple. At the top there is one external incoming label, and at the bottom there is one external outgoing label.

Let e_x, e_y be the omitted incoming labels at the two middle corners, let t be the top incoming label, let b be the bottom outgoing label, and let f_x, f_y be the omitted middle outgoing labels. There are $3^6 = 729$ assignments. The closed neighborhood of the square contains exactly twenty vertices: the four corners and their sixteen external neighbors. Retain two constant inside and two constant outside taxa. They exist because the varying labels consist of three incoming labels, x, y, and three outgoing labels. This gives a twelve-taxon check preserving the twenty vertices and the four corner degrees. Every retained vertex is nontrivial, so its indegree and outdegree are at most three.

The independently checked classification is exactly

$$t \notin \{e_x, e_y\}, \qquad b \notin \{f_x, f_y\}. \tag{14.1}$$

It accepts 144 assignments and rejects 585, each by a degree violation or an explicit forbidden witness. All accepted cases are also checked for compatibility. Condition (14.1) says that the incoming label t and outgoing label b occur at all four corners, proving the two claimed full squares. □

14.1 All other full faces on the same coordinate pair

Put $K = A \setminus \{a_0, a_1, a_2\}$. Up to separate permutations of the incoming and outgoing triples there are four stencils. Their parameters are

type	(e_x, e_y, t)	(b, f_x, f_y)
I	(a_0, a_0, a_1)	(b_0, b_1, b_1)
II	(a_0, a_0, a_1)	(b_0, b_1, b_2)
III	(a_0, a_1, a_2)	(b_0, b_1, b_1)
IV	(a_0, a_1, a_2)	(b_0, b_1, b_2).

The independent permutations are valid: the incoming and outgoing taxa occupy disjoint sets, and the two middle corner positions are kept fixed throughout.

Lemma 14.3. *Suppose $|A|, |X \setminus A| \geq 11$ and $S_{xy}(A)$ is a six-square. For any other full face $S_{xy}(B)$, its base belongs to*

$$\{\varnothing, A - t, A, A + b\} \ \cup \ \big\{\{a_0\} : e_x = e_y = a_0\big\} \ \cup \ \big\{\{b_1\} : f_x = f_y = b_1\big\}.$$

The last two singleton choices occur only in the corresponding equal-omission stencils.

Proof. The twenty old vertices use the eight varying taxa $a_0, a_1, a_2, x, y, b_0, b_1, b_2$. The constant inside set K and constant outside set $L = X \setminus (A \cup \{x, y, b_0, b_1, b_2\})$ each have size at least three. The root belongs to L. Since a foreign base excludes x, y, it has 64 possible patterns on the other six varying taxa.

Partition the constant taxa into the four categories $K \cap B$, $K \setminus B$, $L \cap B$, and $L \setminus B$. Their multiplicities may be saturated at two: a forbidden witness requires at most two taxa of any one row pattern. Exactly twenty saturated profiles are possible under $|K|, |L| \geq 3$ and $r \in L \setminus B$. For each stencil this gives $64 \cdot 20 = 1280$ cases. The cases that survive are as follows. Here integer masks use the varying coordinates in the order $(a_0, a_1, a_2, x, y, b_0, b_1, b_2)$; the foreign face then adds the x, y bits to the displayed base.

type	bases retaining all of K and no L	bases retaining neither K nor L
I	$5, 7, 39$	$0, 1, 64$
II	$5, 7, 39$	$0, 1$
III	$3, 7, 39$	$0, 64$
IV	$3, 7, 39$	0.

All other category profiles are rejected. In particular, there is no freedom to choose a proper nonempty subset of K, or to add any constant outside taxon.

Of the 5120 total cases, 5100 have explicit forbidden witnesses and twenty survive. The verifier reconstructs the old stencil and foreign face independently, checks every witness on its six specified taxa, and checks compatibility of every surviving case. It also reconstructs the saturated category coverage without using the classification producer. No solver is used for these checks. Reading the three large masks as $A - t, A, A + b_0$ and the small masks as the stated empty or singleton sets proves the lemma. □

Theorem 14.4. *For each pair of nonroot coordinates x, y, there is at most one six-square $S_{xy}(A)$ whose base has both sides of size at least eleven. Therefore the number of these squares is at most $\binom{n-1}{2}$, and their vertices together number at most $4\binom{n-1}{2}$.*

Proof. Fix one such six-square. Lemma 14.3 lists every possible base of a full face on its coordinate pair. Empty or singleton bases cannot support a six-square: their bottom corner has indegree at most one and outdegree at most three.

If $S_{xy}(A-t)$ were also a six-square, the lower extension in Lemma 14.2 would force a full face with base $(A-t)-u$, for some $u \in A-t$. This base has size at least nine. It is smaller than all three large permitted bases and larger than every small permitted base, a contradiction. The size assumptions for the extension lemma hold at $A-t$. Similarly, a six-square based at $A+b$ would force a full face at $A+b+v$, for a new outgoing label v. This is larger than all permitted large bases, and again the extension lemma applies. Only A can be the base of a six-square. Counting coordinate pairs and their four corners proves the bounds. □

Corollary 14.5. *Deleting at most $4\binom{n-1}{2}$ nontrivial splits eliminates all six-squares with bases having both sides of size at least eleven. The loss of edges is at most $24\binom{n-1}{2}$. After this deletion, further split deletions cannot create a new sufficiently balanced six-square.*

Proof. Delete the corners of the squares counted by Theorem 14.4. Each deleted corner is nontrivial and has degree six before deletion. Subsequent deletions only lower degrees, and every nontrivial vertex already has degree at most six. Thus a new degree-six square cannot be created. □

The solver-free verifier reconstructs all stencils and saturated profiles and checks the explicit witnesses and survivors. Its data, program, and successful record are:

`six_square_stencil_classification.json`
`six_square_faces_classification.json`
`verify_six_square_structure.py`
`six_square_structure_verification.json`.

15 Cubes with degree-six bottom and top have quadratic total size

Consider a full cube

$$Q(B;x,y,z) = \{B \cup R : R \subseteq \{x,y,z\}\}, \qquad x,y,z \notin B,$$

whose bottom B and top $A = B+x+y+z$ both have degree six. The other six cube vertices may have smaller degrees. Call this cube balanced when both sides of every corner have size at least eleven. The root is omitted from every member, so all its axis labels are nonroot taxa.

Theorem 15.1. *For each coordinate pair $\{x,y\}$, at most two balanced cubes with degree-six bottom and top use x,y as axes. Consequently, if N_{66} is the total number of such cubes, then*

$$N_{66} \le \frac{2}{3}\binom{n-1}{2}.$$

They can all be eliminated by deleting at most $\frac{2}{3}\binom{n-1}{2}$ vertices and at most $4\binom{n-1}{2}$ incident edges. No new balanced cube of this type is created by the deletion.

The proof combines an explicit finite foreign-face classification with a short counting argument. The finite classification is checked without a satisfiability solver.

15.1 The fourteen forced members

Fix one such cube. Its bottom has incoming labels $P = \{p_0,p_1,p_2\} \subset B$ and outgoing labels x,y,z. Its top has incoming labels x,y,z and outgoing labels $Q = \{q_0,q_1,q_2\} \subset X \setminus A$. The nine named taxa are distinct. Retain the eight cube corners, the three predecessors $B-p_i$, and the three successors $A+q_i$. These are fourteen distinct members.

For the coordinate pair x,y, write T for the base of any other full face $\{T, T+x, T+y, T+x+y\}$ in the family. Thus $x,y \notin T$, and T omits the root.

Lemma 15.2 (Foreign-face restriction)**.** *Every such base belongs to one of the following two finite lists:*

$$\mathcal{B}_{\text{large}} = \{B - p_i : 0 \le i < 3\} \cup \{B, B + z\} \cup \{B + z + q_i : 0 \le i < 3\},$$
$$\mathcal{B}_{\text{small}} = \{R \cup S : R \in \mathcal{P}_1,\ S \in \mathcal{Q}_1\},$$

where $\mathcal{P}_1 = \{\varnothing, \{p_0\}, \{p_1\}, \{p_2\}\}$ *and* $\mathcal{Q}_1 = \{\varnothing, \{q_0\}, \{q_1\}, \{q_2\}\}$*. In particular, a face base of size at least three must lie in the eight-member list* $\mathcal{B}_{\text{large}}$*.*

Finite classification with uniform normalization. Outside the nine named taxa, partition the ambient set by membership in B and T. Let their multiplicities be

$$(c_{11}, c_{10}, c_{01}, c_{00}).$$

The first two classes together have size $|B|-3 \ge 8$. The last two together have size $|X \setminus B|-6 \ge 5$; the stated cube balance actually gives a stronger bound. The root belongs to the last class, so $c_{00} \ge 1$. For any four retained columns each class has one row pattern. A forbidden witness needs at most two copies of any row pattern. Truncating every multiplicity at two therefore preserves whether a forbidden witness exists.

The possible truncated profiles, allowing all nonnegative multiplicities with each of the two sums at least three and $c_{00} \ge 1$, form a set of twenty profiles. This deliberately includes every profile arising from the source above. The foreign base has seven independent membership choices on the named taxa $P \cup \{z\} \cup Q$, hence 128 possibilities. The complete test consists of $20 \cdot 128 = 2560$ cases.

Normalize $P = \{0, 1, 2\}$, $x = 3$, $y = 4$, $z = 5$, $Q = \{6, 7, 8\}$, with the named part of B equal to 7. The fourteen named column masks are

$$3, 5, 6, 7, 15, 23, 31, 39, 47, 55, 63, 127, 191, 319.$$

Adding a foreign face, the only surviving pairs of a named base mask and truncated profile are

Named base masks	Profile
$3, 5, 6, 7, 39, 103, 167, 295$	$(2, 0, 0, 2)$
$\{a + b : a \in \{0, 1, 2, 4\},\ b \in \{0, 64, 128, 256\}\}$	$(0, 2, 0, 2)$

There are 24 surviving cases. The first row keeps every unnamed taxon of B and adds none outside B; it is exactly $\mathcal{B}_{\text{large}}$. The second row keeps no unnamed taxon of B and adds none outside; it is exactly $\mathcal{B}_{\text{small}}$.

Every one of the other 2536 cases has an explicit four-column, six-taxon witness, stored in `opposite_six_cube_faces_classification.json`. The independent program `verify_opposite_six_cube_structure.py` reconstructs all cases and checks the four columns, both anchor rows, all four private rows, and coverage of every excluded case. It also checks that a certificate uses no more than its truncated multiplicity from any unnamed class. Thus a witness obtained after expanding a saturated class to three copies still applies when that class has only two available taxa. Finally it checks compatibility of all 24 survivors by a separately implemented four-column obstruction checker. The audit passes. Truncation and the exhaustive certificates give the asserted ambient-independent restriction. □

15.2 At most two cubes on a coordinate pair

An additional balanced cube on x, y has two x, y-face bases L and $L + w$, differing by its third axis w. Both bases have size at least eleven, so both lie in $\mathcal{B}_{\text{large}}$. The inclusion edges of that eight-vertex Hamming graph are exactly

$$B - p_i \longrightarrow B, \qquad B \longrightarrow B + z, \qquad B + z \longrightarrow B + z + q_i.$$

There are seven candidates. The middle edge is the original cube. At most one of the three lower-leaf edges can support another cube: two choices $p_i \neq p_j$ would give four incoming neighbors at their common degree-six top $B + x + y$, with labels x, y, p_i, p_j. Similarly, two upper-leaf choices would give four outgoing neighbors at their common degree-six bottom $B + z$.

It remains to exclude one lower-leaf cube and one upper-leaf cube simultaneously. Suppose they both occur, and write p for the chosen lower label, with $P \setminus \{p\} = \{r, s\}$. The lower cube makes $C = B - p$ degree six, with its three outgoing labels exactly p, x, y. In particular $C + z$ is absent. The upper cube makes $D = B + z$ degree six. Every incoming neighbor $D - t$ other than B must have $t \in P$. Indeed, apply Lemma 9.1 at B to $D - t = B - t + z$: the only removed label is t, and the two needed outside anchors exist because the source is balanced. Since $D - p = C + z$ is absent, the other two incoming neighbors of D must be $D - r, D - s$.

The four present members

$$B - p + x, \quad B - p + y, \quad B - r + z, \quad B - s + z$$

are forbidden: taxa p, z are equal anchor rows, and taxa y, x, s, r, in that order, are the four required private rows. Explicitly the anchor row is 0011, and the private rows are $0100, 1000, 1110, 1101$. This contradiction excludes simultaneous lower and upper additions. There is therefore at most one cube besides the original one, proving the bound two.

15.3 Global count and deletion

Each cube has three coordinate pairs. Summing the bound two over the $\binom{n-1}{2}$ nonroot pairs gives

$$3N_{66} \leq 2\binom{n-1}{2}.$$

Delete the bottom of every counted cube. At most N_{66} vertices are deleted, each of original degree six, so at most $6N_{66} \leq 4\binom{n-1}{2}$ edges are lost. Deleting vertices cannot create a new full cube. It also cannot create a new balanced degree-six vertex: every balanced vertex already had degree at most six, and its degree only decreases. Any putative new cube of the counted type would therefore have existed before deletion, with its bottom among those deleted. This proves Theorem 15.1.

16 Physical boundary membership and its sound relaxation

Graph distance can exceed Hamming distance beyond the retained radius-four ball. In particular, a physical neighbor of a level-four vertex may have Hamming distance three but graph distance five. Its membership must not be identified with membership in the retained prefix. The following separate flags cover all such possibilities.

16.1 Correct physical-neighbor variables

Let u_C mean that C belongs to the retained radius-four prefix. For each old-coordinate candidate neighbor B of a boundary owner, introduce a separate flag e_B for physical membership in the full source family. Impose

$$u_B \implies e_B \quad \text{when } B \text{ belongs to the prefix domain.}$$

For every candidate prefix vertex C at level at most three with $|B \triangle C| = 1$, impose

$$e_B \wedge u_C \implies u_B,$$

where u_B is false if B is outside the prefix domain. This implication is valid because such a selected physical neighbor would have graph distance at most four. In the absence of a selected protected neighbor, e_B is allowed to be true while u_B is false. This includes Hamming-distance-three boundary detours.

The directed degree capacity uses e_B, not u_B:

$$d^{\pm}(D) \le \sum_{\substack{x \text{ an active old slot} \\ x \text{ has sign } \pm}} e_{D\triangle\{x\}} + 3\zeta_D^{\pm}.$$

In the implementation, old slots remain in the sum and the guard $u_D \wedge e_{D\triangle\{x\}} \Rightarrow \mathrm{active}(x)$ excludes inactive ones when the owner is present. The virtual flag $\zeta_D^{\pm}$ permits neighbors on taxa constant on the entire prefix.

Three selected prefix members and a proposed physical neighbor with a forbidden witness force $e_B = 0$. For a constant-class neighbor, the corresponding witness forces $\zeta_D^{\pm} = 0$. Its untouched constant representatives may be chosen separately for each witness, avoiding the taxon being toggled. Each constant class has at least four physical taxa in the sixteen-balanced source, so two such representatives always remain available.

Lemma 16.1 (Soundness of the corrected capacity relaxation)**.** *Every original compatible family with a sixteen-balanced degree-six center admits assignments to the corrected boundary-capacity flags consistent with its actual full directed degrees, after the radius-four normalization.*

Proof. Select exactly the members of the actual radius-four ball. For an old-coordinate candidate, set e_B according to actual full-family membership. Such a neighbor changes only one mapped old taxon, leaving all other taxon entries fixed. Any selected protected neighbor forces its distance to be at most four, proving the guarded prefix implication. For every present boundary owner, set $\zeta_D^{\pm}$ true when it has a full-family neighbor using an unmapped constant taxon of that sign. The old-coordinate sum counts all other neighbors, and at most three neighbors of either sign can belong to the constant class. This proves the degree-capacity inequalities.

A forbidden witness involving three selected prefix members and a purported old-coordinate neighbor excludes that physical neighbor in the original family. The same argument works for every representative of a constant class, using two untouched representatives for repeated rows. Hence every learned exclusion is satisfied. Activity guards discard nonphysical dummy slots. This constructs the required assignment without assuming that every Hamming-close member lies in the retained graph ball. □

17 Local proximity and completion of the strict bound

17.1 The local statement after the additional quadratic pruning

Theorem 17.1 (Computer-assisted radius-two proximity)**.** *Let F be a compatible rooted family in which every member has both sides of size at least sixteen. Suppose every second link is a path system, there is no square with four degree-six corners, and there is no full three-cube whose bottom and top both have degree six. Then every degree-six vertex A has a vertex B with*

$$d_F(A, B) \le 2, \qquad C_2(B) \ge 1.$$

Here the charge is recomputed in F using the two rules of Section 4.

Reduction to the finite model. Suppose a counterexample exists. All final charges are nonnegative integers by Proposition 4.1. Thus every selected vertex in the radius-two ball has charge zero. A vertex of degree at most three retains positive charge: its initial charge is at least two; a stage-one donor loses at most one, except that a degree-two vertex can lose two and still retains

one; a stage-two donor does not donate in stage one and loses at most one. Consequently the entire radius-two ball has degrees four, five, or six.

Retain the complete graph ball of radius four about A. Section 9 proves that graph and Hamming distances agree inside this retained ball, and that at most three labels of each sign first appear at any of its four levels. At most twelve incoming and twelve outgoing labels vary. Since both sides of A have size at least sixteen, at least four taxa of each constant sign remain. Keep two constant representatives of each sign, including the root on the outside.

This gives the 28-coordinate prototype with center 16383 and exactly 2403 potential members in levels

$$1,\ 6,\ 33,\ 236,\ 2127.$$

Deleting constant rows preserves all differences between retained members, hence their graph and every degree through radius three. Full degrees at level four are represented by unknown incoming and outgoing counts, each bounded by three. These counts need not be realized by a constructed extension in the finite model.

Path second links imply that two full faces on the same coordinate pair have nested bases: the two constant inside and two constant outside taxa witness their common intersection and common outside. The 2343 potential full faces give 12726 incomparable pairs, which cannot both occur. The square and opposite-cube hypotheses exclude the corresponding selected degree-six configurations; there are 356 potential three-cubes in this domain. The local completion and new-label restrictions of Lemma 9.3 apply at every retained degree-six vertex through level three, whose full star is retained. These are necessary conditions on the source.

We next describe the charge constraints and the finite case coverage, including the treatment of exterior owners. □

17.2 Six central shapes, without reversing the rooted graph

Continue the counterexample reduction in the preceding proof. Write $I = \{a_0, a_1, a_2\}$ and $O = \{b_0, b_1, b_2\}$ for the six labels at A. The swap graph has edge (i, j) when $A - a_i + b_j$ occurs. Each incoming neighbor has outdegree one plus its row degree, and each outgoing neighbor has indegree one plus its column degree, by the completion lemma. Hence all row and column degrees are at most two. A four-cycle gives a positive-charge rectangle corner within distance two, by Theorem 5.3, so it is excluded.

If A is hard, a perfect matching makes all rows and columns nonempty, so there is no first-stage gain at A. Its charge zero requires exactly one low-degree partner. That partner lies in the radius-two ball, where degrees are at least four, and so has degree exactly four. Its isolated swap edge leaves a two-by-two graph. The absence of a four-cycle and the perfect matching give only a matching or a matching with one additional edge. In the matching case, permute the labels to put its unique degree-four partner at $(0, 0)$. The other two partners have degree greater than four. In the extra-edge case, the unique isolated edge is put at $(2, 2)$.

If A is nonhard, it has no second-stage gain. Charge zero therefore requires exactly one first-stage donor, equivalently one empty row or column in total. Row and column bounds and the absence of a four-cycle leave the following six canonical graphs altogether:

Case	Edges
Matching	$00, 11, 22$
One extra edge	$00, 11, 22, 01$
Star and edge	$10, 11, 22$
Five-vertex path	$10, 11, 21, 22$
Transposed star and edge	$01, 11, 22$
Transposed path	$01, 11, 12, 22$

Here ij denotes (i, j). The last two cases are included explicitly, so no change of root or reversal of the full family is needed. Independent enumeration of all 512 three-by-three graphs gives class sizes $6, 36, 18, 18, 18, 18$ and checks all marked-partner normalizations. The hand reductions justify imposing their disjunction on every source counterexample.

17.3 A matching-free characterization of critical donations

Lemma 17.2. *Let D have degree six, with its full star retained, and let $P = D - a + b$, $L = D - a$, $H = D + b$ all be present. Suppose*

$$d^+(L) = d^-(H) = 2.$$

Then P is a partner of a hard D if and only if every incoming and outgoing neighbor of D occurs on some swap edge. Accordingly, when $d(P) \leq 4$, the latter condition characterizes its second-stage donation to D.

Proof. The completion lemma puts all successors of L among D and its swap row, and all predecessors of H among D and its swap column. The displayed degrees make the edge corresponding to P isolated in both its row and its column. A three-by-three bipartite graph containing an isolated edge has a perfect matching exactly when all six vertices have incident edges. Indeed, remove the isolated row and column. A two-by-two graph with no isolated vertex has a perfect matching. Restoring the isolated edge gives the required matching, containing P. The converse is immediate. □

The 512-graph checker separately verifies this equivalence in all 144 isolated-edge instances. Actual-family checks are recorded in `donation_coverage_verification.json`.

For an owner at distance at most two, its entire star and all swaps lie in the retained ball. The first encoding computes its perfect-matching status directly. The second, separately written encoding uses Lemma 17.2 for each possible donation instead. Both use the actual charge equation

$$d(v) + \text{losses}(v) - \text{gains}(v) = 5$$

at every selected vertex of levels zero, one, and two. First-stage transfers are determined by the full directed degrees. A low-degree partner in that ball can have an owner at distance three or four. Its donation is allowed through an optional exterior-owner flag, subject to the full face, directed-degree identities, degree-six owner, and unique ownership. Requiring a full exterior matching is omitted. Every actual donation satisfies these weaker conditions, so this omission enlarges the finite model.

Boundary full degrees satisfy the corrected physical capacity relaxation of Lemma 16.1. In particular an inward Hamming step is allowed to reach a graph-distance-five physical member, and source membership is never identified with prefix membership without the protected-adjacency guard.

17.4 The finite inconsistency and its scope

Completion of the proof of Theorem 17.1. The frozen data package `proximity_certificate_package.json` contains 3892 universal six-taxon prohibitions and 680 individual-neighbor certificates. Each prohibition fixes six distinct active taxa, two anchors, and four private taxa, and forbids simultaneous presence of the four restricted splits in (2.1). It is an instance of compatibility, not an assumption learned from a possibly invalid boundary assignment. The neighbor certificates each specify three prefix members and one proposed physical or constant-class neighbor, with explicit anchor and private rows. Activity guards and the availability of two untouched constant representatives make them necessary in arbitrary ambient size. Section 16 proves their lifting.

The corrected combined model from `research_cluster_relax_radius4.py`, using direct matching tests, returns UNSAT for the disjunction of all six cases. Its record has boundary encoding `physical_membership_with_protected_adjacency` and `canonical_six_cases` true. The separately written `replay_proximity_donation_model.py` reconstructs the domain by enumerating label subsets, uses the matching-free donation criterion, checks the neighbor certificates directly, and likewise returns UNSAT for all six cases together. The frozen-package replay stores its input hash and solver version; the full finite formula is also exported as SMT-LIB. Only the physical-membership boundary encoding is used as unsatisfiability evidence.

The frozen computation uses Z3 5.1.0.0 and is reproduced, without reading the exploratory search files, by

```
python replay_proximity_donation_model.py -final-certificate
```

Its completed record is

`proximity_donation_model_forced_canonical_capacity_physical_boundary_frozen_replay.json`.

The solver-free `verify_strict_exponent_certificate.py` checks every data clause and neighbor witness, both successful corrected runs, the frozen input and source hashes, and the final arithmetic. Its successful record is `strict_exponent_certificate_verification.json`.

Every genuine counterexample supplies a satisfying assignment: select its entire normalized radius-four ball; assign the actual full directed degrees; choose its actual donations and exterior ownership flags; set physical-membership flags from its full source family and constant-class flags from its actual remaining neighbors. The normalization, central case coverage, charge equations, and the corrected capacity soundness lemma show that each asserted condition is satisfied. Every universal forbidden clause follows from its compatibility. This contradicts the finite UNSAT result. Thus no source counterexample exists, proving the theorem. □

This is a computer-assisted proof using Z3, with separate encodings, solver-free graph and witness checks, and a hand source-to-model reduction. No independently checked formal SAT proof certificate is asserted. A feasible partial model would not by itself be a genuine counterexample; the proof uses only the necessary source-to-model direction and unsatisfiability.

17.5 Quadratic pruning and the global edge saving

Start with any original system and fixed root. First delete the smaller-side-at-most-fifteen boundary using Lemma 7.2. Next perform path-link pruning, remove the degree-six squares, remove the bottoms of the balanced cubes with degree-six bottom and top, and finally perform iterative good-cube pruning. The total number of deleted vertices is at most

$$n+1+\binom{n}{2}\left(\sum_{j=0}^{13} j!3^j+13144\right)+\frac{38}{3}\binom{n-1}{2}=O(n^2).$$

All deletions after the boundary are nontrivial vertices, of original degree at most six. Empty and trivial boundary vertices have total incident degree $O(n^2)$. Hence the total edge cost is also $O(n^2)$.

Call the remaining family F^*. Pure split deletion preserves the path-link property. It cannot create a new degree-six balanced vertex, square, or cube of the excluded types. Newly good cubes are handled by the iterative pruning theorem, which charges each transverse pair at most once. Thus F^* satisfies the local proximity theorem, contains no good cube, and every member has both sides at least sixteen.

Recompute its charges. Let p be its number of positive-charge vertices and H its number of degree-six vertices. Assign every six-vertex to a positive vertex within distance two. Since the maximum degree is six, any radius-two ball has at most $1 + 6 + 6 \cdot 5 = 37$ vertices. Therefore

$$H \le 37p.$$

Proposition 13.3 gives

$$V^* \le 938p + 23447H \le (938 + 37 \cdot 23447)p = 868477p.$$

All charges are nonnegative integers and their sum is $5V^* - 2E^*$. Consequently

$$5V^* - 2E^* \ge p \ge \frac{V^*}{868477}, \qquad E^* \le \left(\frac{5}{2} - \frac{1}{1736954}\right) V^*.$$

Transferring back loses only $O(n^2)$ edges, while $V^* \le V$. This proves the unrestricted edge inequality in Theorem 1.1.

17.6 Standard reduction on the original unrestricted class

Let $f(n)$ be the largest size of a compatible split system on n taxa, including the possible empty split. For a nonroot taxon x, restricting a rooted family of size V can merge only pairs $B, B + x$. Hence

$$|E_x| = V - |\{B \setminus \{x\} : B \in F\}| \ge V - f(n-1),$$

where E_x is the edge set with label x. Summing over the $n - 1$ nonroot labels and applying the unrestricted edge inequality gives, for $n - 1 > \alpha$,

$$(n - 1 - \alpha) f(n) \le (n-1) f(n-1) + Cn^2.$$

The recurrence is applied to the original class, which is hereditary under taxon deletion. No hereditary assertion about the pruned path-link class is needed.

For fixed sufficiently large n_0, set

$$g(n) = \prod_{j=n_0+1}^{n} \frac{j-1}{j-1-\alpha}.$$

Taking logarithms gives $g(n) = \Theta(n^\alpha)$. Dividing the recurrence by $g(n)$ yields

$$\frac{f(n)}{g(n)} \le \frac{f(n-1)}{g(n-1)} + O(n^{1-\alpha}).$$

Since $\alpha > 2$, the error series converges. Thus $f(n) = O(n^\alpha)$, completing the proof of Theorem 1.1.

A Computational dependencies and reproducibility

The text above is a single exposition of the proof, with no references to omitted research-note sections. The finite programs and data remain part of the computer-assisted argument; a printed document alone does not replace their verification. The facts attributed to [1] are the published inputs.

For the final proximity computation, run from the research directory:

```
python replay_proximity_donation_model.py -final-certificate
python verify_strict_exponent_certificate.py
```

The first command reconstructs the frozen finite model and invokes Z3; the second checks its input witnesses, recorded successful runs, hashes, supporting records, and the final arithmetic without a solver. Checking a recorded solver outcome is not a formal verification of unsatisfiability. No DRAT, LRAT, or independently checked SMT proof certificate is asserted.

The frozen package contains 3892 forbidden-configuration clauses and 680 boundary-neighbor witnesses. The independently constructed formula has 2403 potential prefix members. The recorded frozen replay is UNSAT using Z3 5.1.0.0. Its exact identifiers are:

Package SHA-256:
`fa58d215e75808f34d0e546dd508bd49edc48b99f9d48a69a32a7bfffb082142`

Replay program SHA-256:
`c611d19e31444524aa21481a67e71a3d272d4ee82c62aaf8744042912a44d02c`

SMT-LIB formula SHA-256:
`67ee62db9c581f527c79d88565dd17e7f0e34d50810b9f2581dd8fda6109812b`

The frozen package is `proximity_certificate_package.json`; the formula is `output/proofs/proximity_model.smt2`. The audit record `strict_exponent_certificate_verification.json` also lists the hashes of all supporting computation records. The preceding sections identify the producer, independent verifier, finite domain, and normalized counterexample for each required finite exclusion. Separate implementations share the mathematical normalizations and Z3, and are not independent formal proof checkers.

Remark A.1 (The two-layer grid does not remove the quadratic error). The final example in [1] has $|V| = \Theta(n^2)$ and $|E| = (5/2)|V| - O(n)$. Hence $|E| - \alpha|V| = \delta|V| - O(n)$ is quadratic and can be absorbed in Cn^2. Its regular interior contains the good cubes treated above. The proof does not claim positive charge density on that original grid. It applies its charge-density estimate after the quadratic deletions and then transfers the result back with the quadratic error. This error is consistent with the size exponent because $\alpha > 2$.

References

[1] P. Wu and S. Grünewald, *On the Maximum Size of 2-Weakly Compatible Split Systems*, arXiv:2608.23275v1 (2026). `https://arxiv.org/abs/2608.23275`.